Model Contract Guide

The baseline model includes:

- 4 linear layers
- Width 256
- Depth 4
- SiLU activations
- no output activation

Architecture: input -> 256 -> 256 -> 256 -> C

First objective: MSE

- predict RGB value directly from embedded coordinates

Second objective: CICFM

- predict velocity from coordinate, time, and interpolated value

Frozen Model Contract

coordinates: [B, N, D]
targets: [B, N, C]
features: [B, N, D + 2 * D * F]
prediction: [B, N, C]

MSE Baseline Semantics
gamma(x) = concat(
x,
sin(pi * 2^0 * x), cos(pi * 2^0 * x),
sin(pi * 2^1 * x), cos(pi * 2^1 * x),
...
sin(pi * 2^(F-1) * x), cos(pi * 2^(F-1) * x)
)

y_hat = mlp(gamma(x))
loss_mse = mean((y_hat - y)^2)

CICFM Semantics
z0 = random source value, shape [B, N, C]
z1 = target value, shape [B, N, C]
t = uniform time in [0, 1], broadcast to [B, N, 1]

zt = (1 - t) * z0 + t * z1
target_velocity = z1 - z0

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model_input = concat(gamma(x), zt, t)
pred_velocity = mlp(model_input)
loss_cicfm = mean((pred_velocity - target_velocity)^2)
```

To Do:

- [] Freeze model and loss contract
- [] Define MLP forward semantics and MSE target
- [] Define CICFM Scaffolding
- [] Verify MSE training logic and backward math
- [] Convert MSE to CICFM (use explicit loss mode)
- [] Debug model behavior and loss transition
- [] Document the final model and loss design